

UG Project Report submitted in partial fulfilment for the award of the
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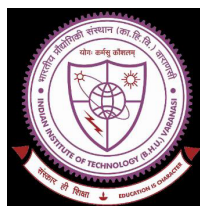
“Urban Transport Accessibility in India: A Comparative Study of PTAL Scores in Bangalore and Varanasi”

By

Abhijeet Kumar Gaurav
(20065001)

Under the Guidance of

Dr. Abhisek Mudgal
Assistant Professor



Department of Civil Engineering

Indian Institute of Technology (Banaras Hindu University)

Varanasi – 221005

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Abstract

Public transport ensures a city's economic and social well-being by facilitating efficient mobility, reducing automobile dependence, and mitigating traffic congestion, and one of the key determinants of a well-functioning public transport system is the level of accessibility it offers. This project evaluates the Public Transport Accessibility Level (PTAL) in Bangalore, emphasising the Bangalore Metropolitan Transport Corporation (BMTC) mode, and extends the analysis to Varanasi's public bus network. Drawing on the PTAL methodology originated by London's Borough of Hammersmith and Fulham and subsequently adopted by Transport for London (TfL), the model is adapted to India's urban context using a grid-based variant: the study area is segmented into 1 km² grid cells, whose centroids serve as points of interest (POIs), and PTAL is computed with respect to bus stops as service access points (SAPs). A Geographic Information System (GIS) mapping workflow (QGIS with Python scripting) is used to represent PTAL levels visually, considering average walk speed and time, distances to public transport stops, and average frequencies of public transport modes. In addition to PTAL mapping, a comparative regression analysis of PTAL score against population was conducted for both cities. The findings show that while both Bangalore and Varanasi have experienced population growth, the relationship between population and PTAL score differs: Varanasi exhibits a weak, statistically significant *negative* relationship ($R^2 = 0.086$), whereas Bangalore exhibits a weak, statistically significant *positive* relationship ($R^2 = 0.018$), reflecting Bangalore's heavier investment in public transport infrastructure and supportive policy. The study concludes by discussing the potential applications of PTAL mapping in enhancing planning practice, including formulating development/master plans with land use-transport integration, prioritising public transport and supporting investment, formulating parking policies, and developing transit-oriented zoning regulations.

Keywords: Public Transport Accessibility Level (PTAL); Accessibility; GIS; QGIS; Bangalore Metropolitan Transport Corporation (BMTC); Varanasi City Transport Service (VCTSL); Regression Analysis; Transit-Oriented Development

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1 Introduction

The effectiveness of public transportation in an urban area is critically assessed using the Public Transport Accessibility Level (PTAL) index. It gauges the ease with which residents can access and utilise public transportation services to reach their desired destinations. PTAL scores are significant for rapidly urbanising cities like Bangalore and Varanasi, which grapple with mounting transportation challenges stemming from population growth, economic development, and infrastructure constraints. This comparative study delves into the PTAL scores of Bangalore and Varanasi, identifying the factors contributing to their differences and drawing insights for enhancing public transport accessibility in these contrasting urban landscapes.

1.1 Background

Bangalore, the capital of Karnataka, stands as India's third-most populous city and a prominent hub for technology and innovation. Its rapid urbanisation trajectory has triggered a surge in transportation demand, placing immense strain on the existing infrastructure. Despite initiatives to enhance public transportation, such as introducing the Namma Metro, traffic congestion remains a persistent challenge, negatively impacting PTAL scores.

In contrast, Varanasi, one of the world's oldest continuously inhabited cities, is renowned for its rich cultural heritage and spiritual significance. Its narrow lanes, ancient architecture, and dense population density present unique challenges for modern transportation infrastructure development. While the city boasts a comprehensive network of rickshaws and bicycle taxis, the overall PTAL score still needs improvement due to the limitations of these modes of transport.

1.2 Necessity for Comparison

A comparative analysis of PTAL scores between Bangalore and Varanasi offers a valuable opportunity to examine the interplay between urban development, historical heritage, and transportation accessibility. By understanding the factors influencing PTAL scores in these contrasting urban landscapes, insights can be derived for improving public transportation systems and enhancing accessibility for residents.

1.3 Demographic Trends and PTAL Scores

Bangalore's population, estimated to reach 12.1 million by 2031, has witnessed exponential growth averaging 3.3% annually. This rapid urbanisation has exacerbated transportation challenges, as the existing infrastructure struggles to accommodate growing mobility demand; the city's PTAL scores reflect this strain, with specific areas facing severe congestion and limited access to public transportation.

Varanasi, with an estimated population of 1.4 million, has grown at a slower pace of 1.5% annually. However, its dense population, concentrated primarily along the Ganges River, poses

unique challenges for public transportation planning and infrastructure development. The city's PTAL scores reflect these challenges, with specific areas experiencing limited connectivity and reliance on traditional modes of transport.

1.4 Public Transport Data: BMTC and VCTSL

The Bangalore Metropolitan Transport Corporation (BMTC) plays a pivotal role in providing public transportation services in the city. With a fleet of over 6,000 buses, the BMTC operates an extensive network of routes, catering to a daily ridership of over 3.5 million passengers. However, despite the BMTC's efforts, traffic congestion and inadequate infrastructure continue to hinder its ability to provide efficient and accessible public transportation services.

Varanasi's public transportation, managed by the Varanasi City Transport Service Ltd. (VCTSL), is a vital mobility hub in this ancient spiritual city. Established in 1980, VCTSL operates over 200 buses, serving more than 1 lakh passengers daily across 20 well-designed routes that ensure easy access to key landmarks and residential areas. Complementing the bus network are ubiquitous rickshaws and bicycle taxis, offering flexibility especially in areas inaccessible by buses. While recent upgrades, such as air-conditioned buses and cashless ticketing, have improved services, challenges persist: narrow lanes and traffic congestion hinder seamless operations, prompting plans for expansion, including electric buses and integration with the proposed metro rail, to create a more accessible and sustainable public transportation system for Varanasi.

1.5 Economic Growth and Transportation Needs

Bangalore's robust IT sector and large workforce have amplified the demand for reliable and accessible public transportation; the city's economic growth hinges on the ability to connect workers to employment opportunities, and efficient public transport is crucial to this goal. However, the current transportation infrastructure needs to improve to meet the demands of the growing economy, leading to congestion and reduced productivity.

In contrast, Varanasi's predominantly informal economy and reliance on traditional modes of transport have limited the impact of public transportation on economic growth. The city's economic activities are primarily centred around tourism and religious pilgrimages, and the existing rickshaw and bicycle taxi networks cater to these needs; however, the need for a modern public transportation system constrains the city's ability to attract new industries and businesses.

2 Literature Review

2.1 Defining Accessibility

Accessibility refers to the ease of reaching a desired destination or service, typically in the context of transportation. It measures the degree to which a particular location is reachable or approachable by different modes of transportation, such as walking, cycling, public transit, or

private vehicles. Several standard definitions have been proposed by accessibility researchers and transportation experts:

1. The ease with which activities can be reached from a point of origin, usually measured in time or distance (Handy, 2002).
2. The degree to which a transport system enables access to destinations and activities, thus promoting land use and travel patterns that sustain social cohesion, environmental integrity, and economic vitality (Banister & Hickman, 2006).
3. The ability of people to reach the goods, services, and activities that they need or desire, regardless of physical or economic barriers (Cervero & Kockelman, 1997).
4. The degree to which transportation systems provide equitable access to opportunities and services, regardless of income, age, gender, or physical ability (Guo & Wilson, 2015).

These definitions highlight the importance of accessibility in promoting sustainable, equitable, and efficient transportation systems that meet the needs of all users.

2.2 Overview of the PTAL Methodology

The Public Transport Accessibility Level (PTAL) methodology is a framework developed by Transport for London (TfL) to assess the accessibility of public transport in London. The PTAL methodology assigns a score to each location in London based on the ease with which residents can access and utilise public transportation services to reach their desired destinations.

2.2.1 Critical Components of the PTAL Methodology

The PTAL methodology considers several factors when calculating PTAL scores, including:

- **Proximity to public transport stops:** the closer a location is to public transport stops, the higher its PTAL score.
- **Frequency of public transport services:** the more frequent public transport services are in an area, the higher its PTAL score.
- **Variety of public transport modes:** the greater the variety available in an area, the higher its PTAL score.
- **Quality of public transport infrastructure:** the condition of bus stops and stations also affects PTAL scores.

2.2.2 PTAL Scores and Urban Planning

PTAL scores are used in urban planning to identify areas with low accessibility and inform public transport investment decisions. For example, areas with low PTAL scores may be prioritised for new bus routes or improved bus stop infrastructure.

2.2.3 Limitations of the PTAL Methodology

Despite its benefits, the PTAL methodology also has limitations. For example, it does not consider factors such as personal preferences or the cost of public transport, and PTAL scores can be misleading in areas with high car ownership, as residents may rely on public transport less than in other areas. Overall, the PTAL methodology remains a valuable tool for assessing public transport accessibility, informing decisions about public transport investment, and tracking progress towards improving accessibility for all residents.

2.2.4 PTAL Score Range

The PTAL score ranges from 0 to 6, each representing distinct public transport access levels within a given area (Table 1).

Table 1: PTAL score bands (London/TfL convention).

PTAL	Range of Index	Description
1a (Low)	0.01–2.50	Very Poor
1b	2.51–5.00	Very Poor
2	5.01–10.00	Poor
3	10.01–15.00	Moderate
4	15.01–20.00	Good
5	20.01–25.00	Very Good
6a	25.01–40.00	Excellent
6b (High)	40.01+	Excellent

2.3 Why the PTAL Score Was Selected for This Study Area

Varanasi, renowned for its historical significance, cultural diversity, and intricate transportation network, is a captivating study area. With a population exceeding 1.2 million, the city presents intricate mobility challenges amid its rich heritage and religious importance. Varanasi’s diverse demographics, amplified by tourism, contribute to the complexity of its transportation dynamics. Additionally, its involvement in government-led initiatives like the Smart Cities Mission allows one to analyse policy interventions and their impact on urban transportation.

In tandem with studying Varanasi, the selection of Bangalore, a city emblematic of rapid urbanisation and burgeoning population, adds depth to this comparative analysis. Bangalore grapples with mounting pressures on its public transport system, which, in turn, poses mobility challenges for its residents amid its cultural diversity.

The study uses the PTAL methodology to comprehensively assess public transport service accessibility in both cities. By scrutinising crucial factors such as walking distances to transport stops, route frequencies, and transport capacity, the PTAL analysis offers insights into the diverse urban transport patterns and complexities prevalent in Indian cityscapes. Furthermore, this comparative study allows for a nuanced understanding of policy impacts on urban transportation,

steering the development of more responsive and efficient transportation systems in both Varanasi and Bangalore.

3 Methodology

3.1 Data Collection

The data collection process for this study aimed to gather comprehensive information regarding Public Transport Accessibility Levels (PTAL) in the Varanasi and Bangalore areas. The primary objective was to calculate the accessibility index and map PTAL scores across both cities to analyse accessibility levels and identify areas for potential infrastructure enhancements. Table 2 summarises the data types and sources used.

Table 2: Data types and sources.

Data Type	Data Source
Varanasi Ward Map	OpenStreetMap (OSM)
Varanasi Bus Stops (GIS file)	Field survey
Varanasi Road Network (GIS file)	OSM
Varanasi 2011 Population Data (ward-wise)	Government site
Varanasi Bus Frequency Data	Field survey
Bangalore Ward Map	OSM
Bangalore Bus Stops (GIS file)	OSM
Bangalore Road Network (GIS file)	Supervisor-provided
Bangalore 2011 Population Data (ward-wise)	Government site
Bangalore Bus Frequency Data	Survey / OSM

The dataset comprises origin IDs, stop IDs, and geographical coordinates (X_POI, Y_POI and X_SAP, Y_SAP), ward-wise population, male population, female population, and the frequency of buses at each bus stop. QGIS and Python scripting were used for data management and analysis.

3.2 The Grid Method

This study departed from the conventional London methodology by adopting a different approach. Instead of the traditional method of identifying discrete points of interest, it segmented the study area into smaller 1 km² grid cells, using their centroids as points of interest (POIs) for PTAL measurement. This modification significantly expedited the analysis process, allowing for quicker insights despite limitations in available data resources.

Varanasi’s distinct urban environment necessitated a re-evaluation of the PTAL assumptions established in London. The city’s prevalent street vendors and inadequate footpaths, which compel pedestrians to walk along roadsides, raised safety concerns and influenced the recalibration of walk speed assumptions; the study revised the walk speed range from 3.5 km/hr to 5 km/hr to better represent the reality pedestrians face. Moreover, increased bus reliability factors were

incorporated to account for Varanasi’s unpredictable traffic conditions and non-adherence to traffic regulations, affecting travel time reliability. These adaptations were crucial in refining the PTAL calculations, aligning the methodology with each city’s unique urban landscape and challenges (Table 3).

Table 3: Model parameters used for Varanasi and Bangalore.

Parameter	Units	Varanasi	Bangalore
Running time of bus	Hour	15	12
Walk speed	km/hr	4	5
Reliability (K)	Min	2.5	2.5
Maximum walk time	Minutes	30	20
Total POIs	—	70	716
Total SAPs	—	143	1683

POI-to-SAP distances were measured using the QGIS tool QNEAT and its OD Matrix option; these parameters were then used to calculate the accessibility index (AI) for each grid cell.

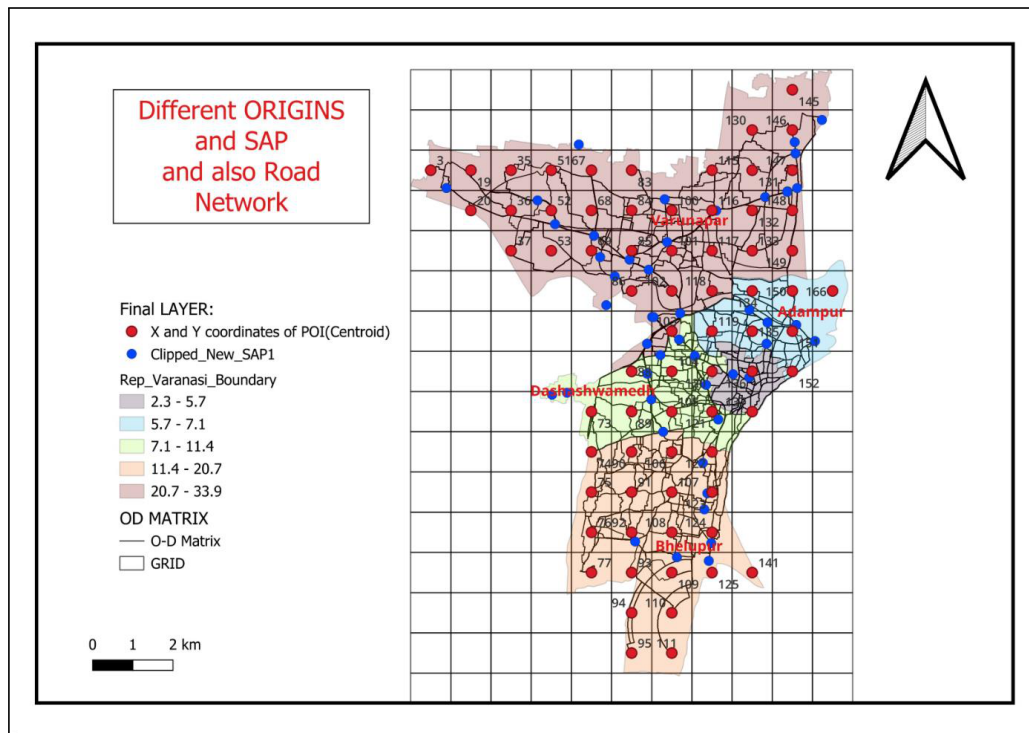


Figure 1: Points of interest (POI, red), service access points (SAP, blue), road network, and OD matrix for Varanasi.

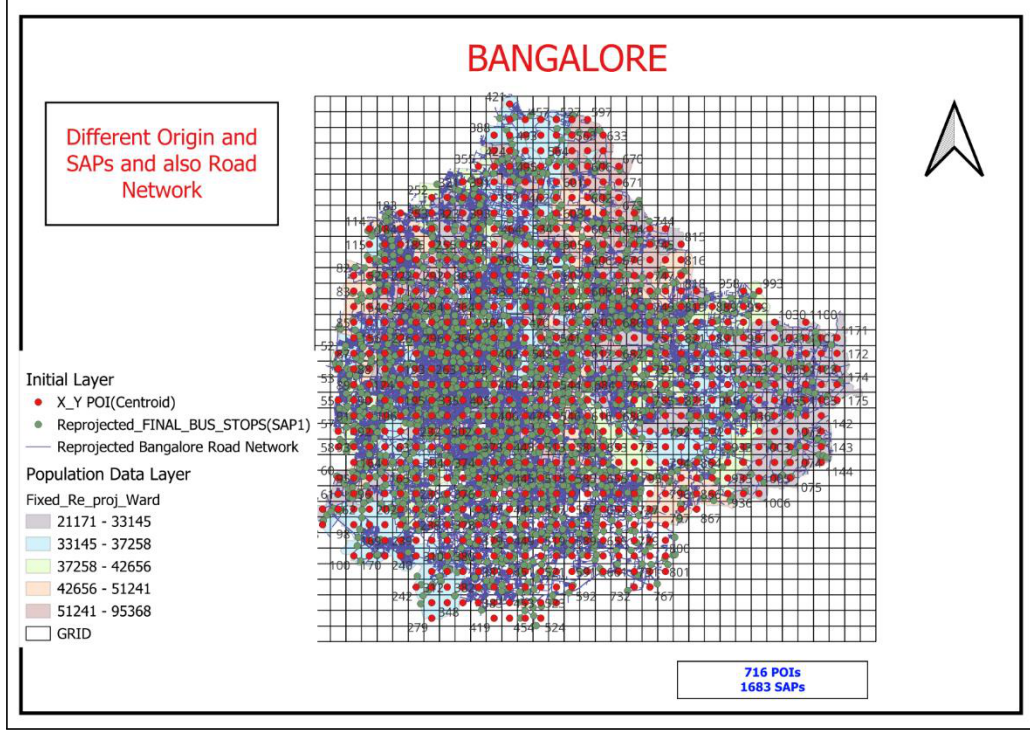


Figure 2: Points of interest (POI, red), service access points (SAP, green), road network, and ward-wise population layer for Bangalore (716 POIs, 1683 SAPs).

3.3 Applying the PTAL Method to the Bangalore and Varanasi Datasets

Step 1: Define points of interest (POI) and service access points (SAP). A POI is a point for which the accessibility level is to be measured with respect to an SAP, which is a public transport stop (bus stop, metro station, etc.); in this study, bus stops serve as the SAPs.

Step 2: Calculate walk access time from POI to SAP. The actual road-network distance from POI to SAP is measured and, assuming the walk speeds in Table 3, the walk time (WT) is calculated. SAPs beyond the maximum walk time are not considered when estimating PTAL for that POI.

Step 3: Identify valid routes at each SAP and calculate the average waiting time (AWT). Valid routes are bus and metro routes during the peak hour (8:00–10:00 AM), and the frequency of services on all such routes during this hour is used in the AWT calculation:

$$AWT = 0.5 \left(\frac{60}{f} \right) + K \quad (1)$$

where AWT is the time between a passenger's arrival at an SAP and the arrival of the desired service, the hourly frequency (f) is halved because the scheduled waiting time (SWT) is estimated as half the headway, and a reliability factor (K) is added to the SWT depending on the transport mode.

Step 4: Calculate the minimum total access time (TAT) for each valid route at each

SAP.

$$\text{TAT} = \text{WT} + \text{AWT} \quad (2)$$

Step 5: Convert TAT into equivalent doorstep frequency (EDF). The principle is to treat access time as a notional average waiting time as though the route were available at the doorstep of the selected POI.

$$\text{EDF} = \frac{30}{\text{TAT}} \quad (3)$$

The reason for dividing 30 (minutes) by TAT is that it re-applies the half-the-headway rule. This is applied twice because the two values have different meanings. In Step 3, frequency is converted into AWT; in Step 5, TAT is converted back into a frequency (EDF). TAT is a number comparable to service frequency but takes into account the additional walk time to reach the stop along with reliability; thus, the half-the-headway rule is applied again to TAT in Step 5 to give the doorstep frequency.

Step 6: Obtain the accessibility index (AI) for each POI. The most dominant route (the route with the highest frequency) is assigned a weighting factor of 1.0; for all other routes, a weighting factor of 0.5 is assigned. For a transport mode (m):

$$(\text{AI})_m = (\text{EDF})_{\max} + 0.5 \sum_{\text{all other routes}} \text{EDF} \quad (4)$$

and the accessibility index for a POI is

$$(\text{AI})_{\text{POI}} = \sum_m (\text{AI})_m \quad (5)$$

Step 7: Map PTAL. The AIs obtained for each POI are allocated to eight PTAL bands as in Table 1 (for Bangalore) and Table 4 (for Varanasi, recalibrated for the smaller AI range observed there). A POI with a value of 0 indicates no access to the public transport network within the given parameters and is left uncoloured on the map.

Table 4: PTAL score bands adopted for Varanasi.

PTAL	Range of Index	Description
1a (Low)	0.01–1.05	Very Poor
1b	1.51–4.00	Very Poor
2	4.51–7.00	Poor
3	7.01–10.00	Moderate
4	10.01–15.00	Good
5	15.01–20.00	Very Good
6a	20.01–30.00	Excellent
6b (High)	<30	Excellent

4 Results and Discussion

4.1 Comparison of PTAL Scores: Bangalore and Varanasi

Figures 3 and 4 present the PTAL score maps for Bangalore and Varanasi respectively, highlighting the areas with high, medium, and low accessibility.

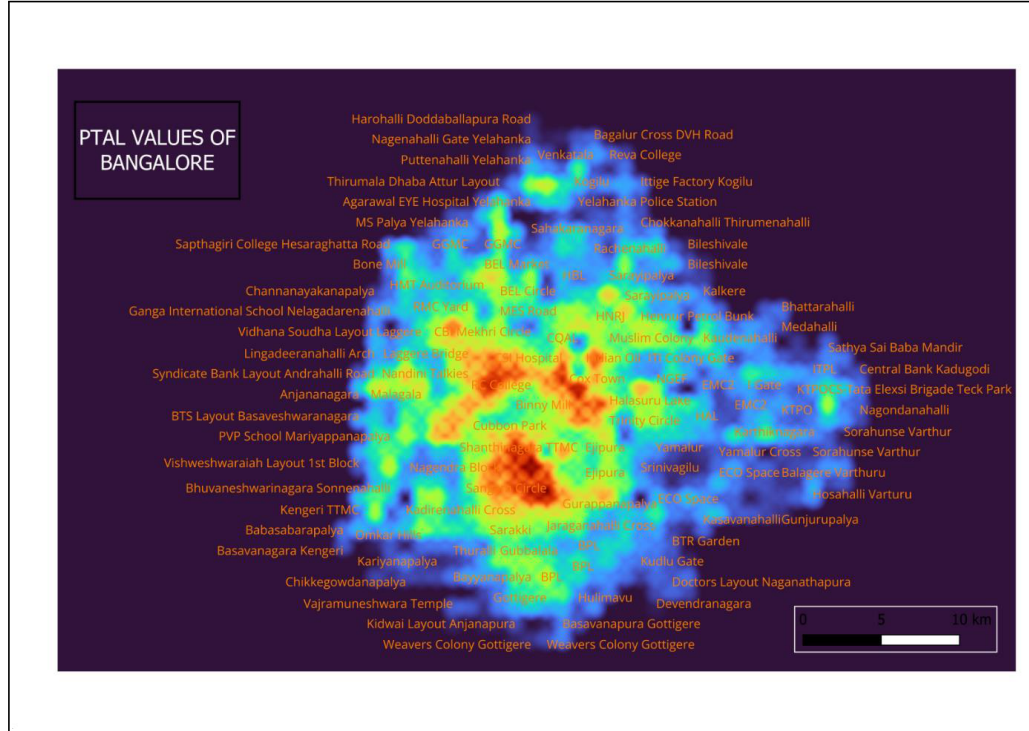


Figure 3: PTAL heatmap of Bangalore, showing accessibility concentrated around the central business district and declining toward the periphery.

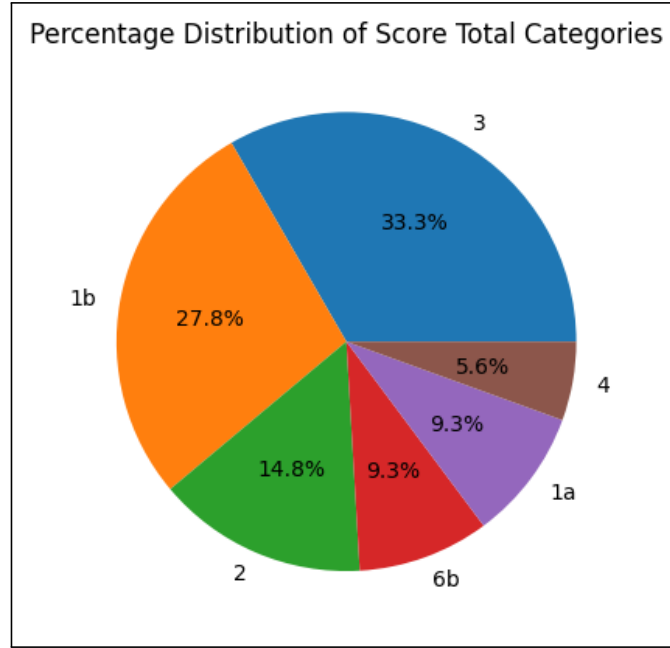


Figure 6: Percentage distribution of PTAL score-total categories, Varanasi.

4.2 Relationship Between PTAL Scores and Urban Characteristics

To discuss the relationship between PTAL scores and population density, Varanasi's grid cells were classified into six categories: Low PTAL with High Population, Moderate PTAL with High Population, High PTAL with High Population, Low PTAL with Low Population, Moderate PTAL with Low Population, and High PTAL with Low Population.

Low PTAL with high population. Several densely populated areas of Varanasi, namely Shivpurwa, Pardepur, Tulsipur, Pahariya, Jolha, and Saraisurjan, combine high population with poor accessibility (Figure 7). In Shivpurwa, limited road width and congestion impede efficient bus manoeuvring, resulting in prolonged travel times, and the absence of exclusive bus lanes compels buses to share road space with other vehicles, causing further delays; overcrowded bus stops further inconvenience commuters. Pardepur's remote positioning on the city outskirts complicates accessibility from the central district, with sparse bus services hindering travel especially during non-peak hours, compounded by poor road conditions. Tulsipur suffers from inadequate infrastructure (bus stops, signage, lighting) and uncoordinated bus schedules that cause prolonged waits and missed connections. Pahariya's hilly topography and narrow streets restrict bus navigation and limit route reach. Jolha experiences heavy peak-hour congestion and inadequate parking that forces buses to stop inconveniently, disrupting flow. In Saraisurjan, an insufficient bus fleet relative to the growing population results in overcrowding and compromised accessibility, with inefficient routing potentially overlooking population distribution and travel patterns.

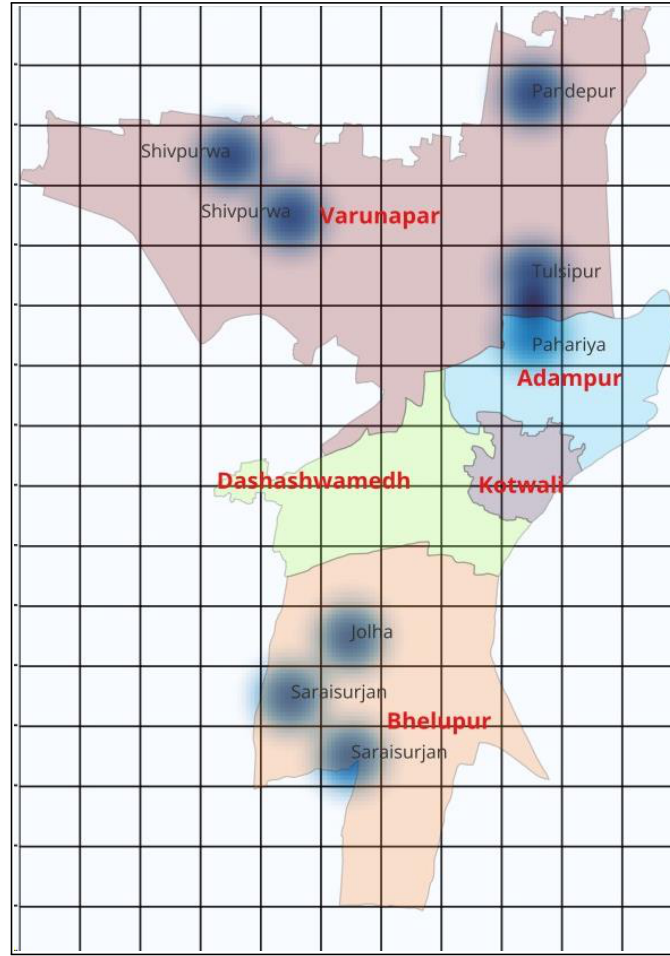


Figure 7: Low PTAL value with high population, Varanasi.

Moderate PTAL with high population. Tarna and Nariya in Varanasi face escalating demands for efficient public transport amid growing populations. While maintaining moderate PTAL, a lack of proactive measures might lead to declining accessibility, posing risks including amplified traffic congestion, diminished economic activity, environmental degradation, and social disparities. The combined populace of this cluster surpasses 200,000, yielding a density of over 30,000 individuals per square kilometre; the transport network comprises buses, rickshaws, and auto-rickshaws, yet insufficient bus frequency and a lack of exclusive lanes result in delays and congestion. Despite these challenges, over 60% of residents rely on public transport for commuting. Left unaddressed, growing populations could intensify congestion and pollution, hinder economic access, and accelerate environmental deterioration; potential solutions include expanding bus service coverage, designating dedicated bus lanes, enhancing road network infrastructure, upgrading bus stop facilities, integrating public transport modes, implementing real-time bus tracking, and employing data-driven decision-making.

High PTAL with high population. Nevada and Tulsipur, two densely populated neighbourhoods, enjoy high PTAL owing to a combination of factors (Figure 8): strategic, centrally located positioning providing easy access to major employment hubs, educational institutions, and commercial centres; an efficient, well-planned road network with dedicated bus lanes enabling

smooth traffic flow; adequately placed and frequent bus stops ensuring easy access to public transport; and seamless integration with rickshaws and auto-rickshaws allowing convenient onward travel.

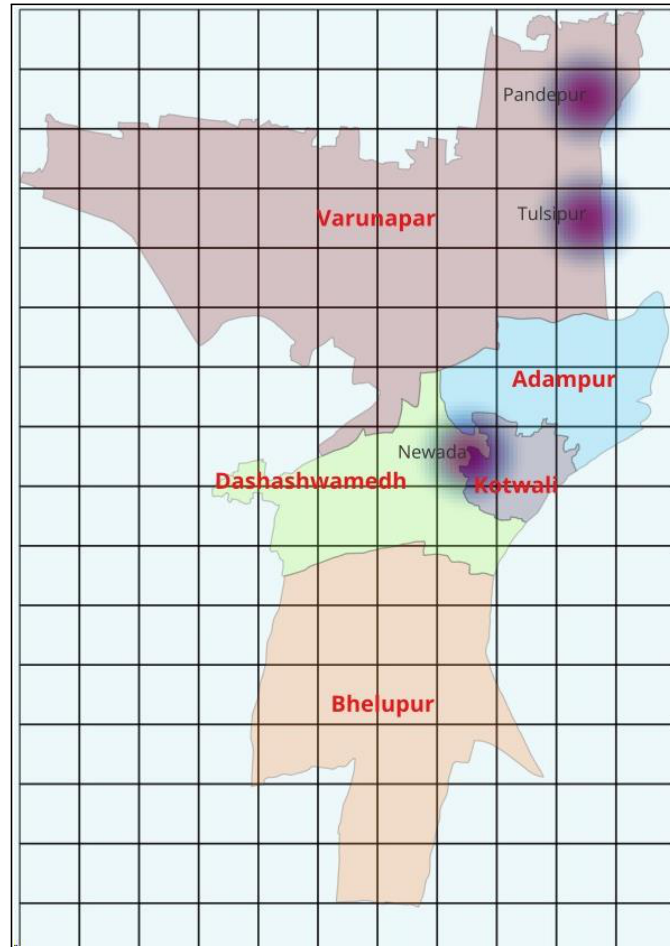


Figure 8: High PTAL with high population, Varanasi.

Corresponding lower-population variants of these categories were also examined (Low, Moderate, and High PTAL with Low Population), confirming that population density alone does not determine PTAL: several low-population grid cells still register poor accessibility due to peripheral location or thin route coverage, while others enjoy high accessibility from proximity to well-served corridors. Figure 9 presents the full set of six PTAL-versus-population subplots used to identify these categories.

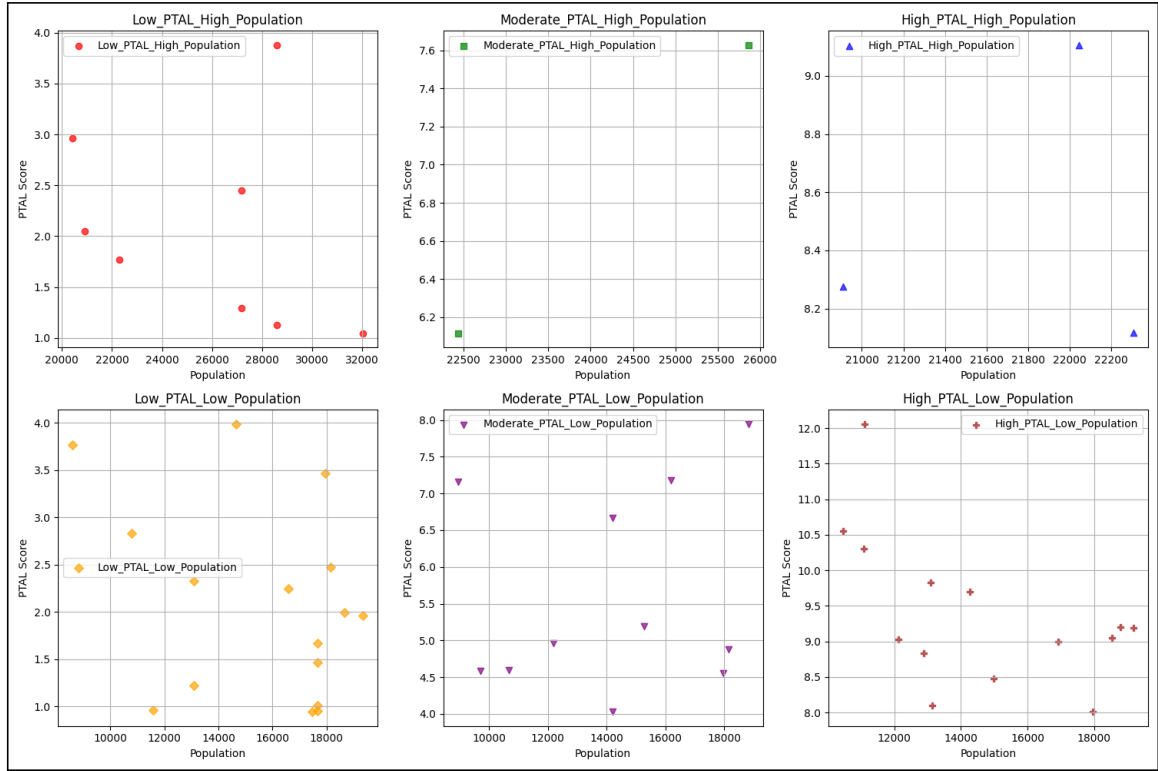


Figure 9: Subplots of PTAL score against population across the six identified categories, Varanasi.

4.3 Regression Analysis: Comparison Between Varanasi and Bangalore

In the comparison of Regression Analysis between the Varanasi and Bangalore datasets, several vital distinctions and similarities emerge, shedding light on their respective public transport accessibility landscapes.

4.3.1 For Varanasi

Figure 10 shows the bubble plot of PTAL score against population for the six identified categories, and Figure 11 shows the fitted OLS regression line.

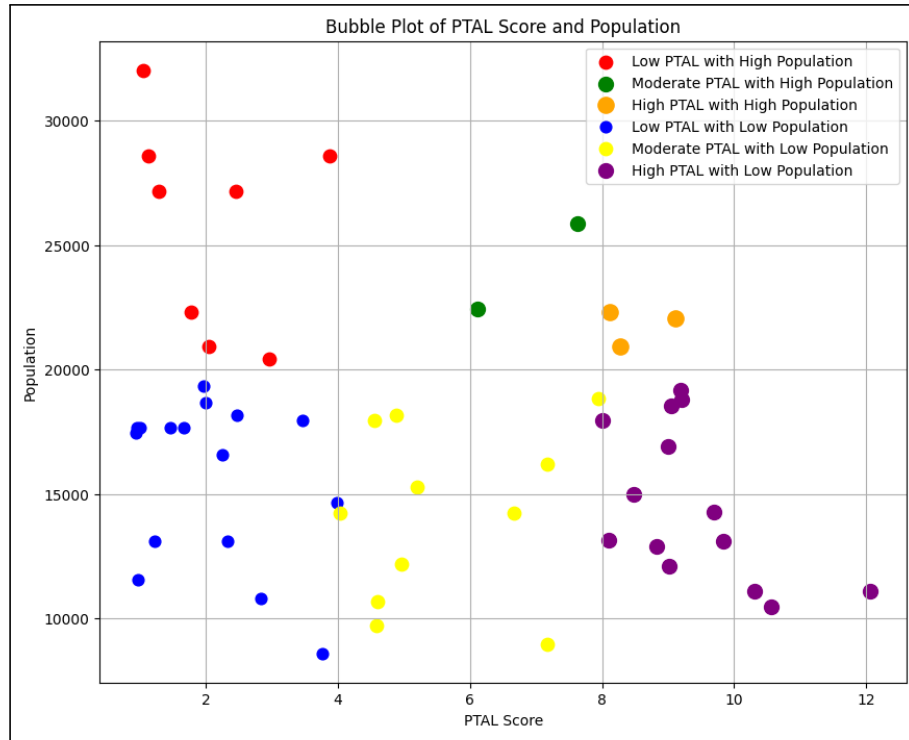


Figure 10: Bubble plot of PTAL score and population, Varanasi.

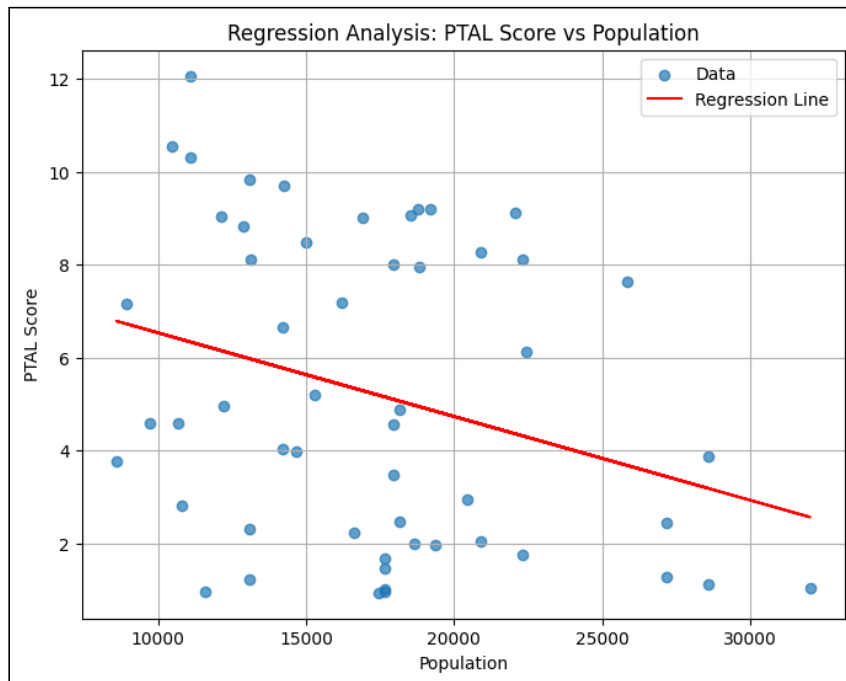


Figure 11: Regression analysis of PTAL score vs. population, Varanasi.

The regression analysis conducted between population and PTAL score for Varanasi yields the following insights:

- **R-squared and adjusted R-squared:** an R^2 of 0.086 suggests that changes in population can explain approximately 8.6% of the variability in PTAL scores; the adjusted R^2 ,

accounting for the number of predictors, stands at 0.068.

- **Significance of the population coefficient:** the coefficient for population is -0.0002 with a p-value of 0.031, suggesting that for every unit increase in population, the PTAL score decreases by 0.0002; the coefficient's p-value is below 0.05, indicating some significance, though the effect size is notably small.
- **Intercept coefficient:** the intercept value is 8.3325, indicating an estimated PTAL score of approximately 8.33 when the population is zero.
- **Model significance:** an F-statistic of 4.888 with p-value 0.0315 suggests the overall regression model is statistically significant at the 0.05 level.
- **Model fit:** the low R-squared value indicates that the model does not fully capture the variability in PTAL scores, and other factors likely play a more substantial role.

The analysis shows a statistically significant, albeit weak, *negative* relationship between population and PTAL score in Varanasi: an increase in population might marginally lead to a slight decrease in PTAL scores. This finding suggests that Varanasi's growing population may pose a challenge in maintaining and improving public transport accessibility, potentially due to overburdened infrastructure (bus routes, stops, and fleet size becoming strained as population grows, leading to overcrowding and reduced accessibility), traffic congestion (increased population density making it more difficult for buses to navigate efficiently), and limited resources (constrained ability to expand the network to meet growing demand).

A declining PTAL trend could also affect Varanasi's cultural heritage: reduced accessibility to cultural sites (temples, ghats, museums) could hinder tourists and residents from fully experiencing the city's offerings; limited mobility could restrict participation in cultural events and gatherings, particularly for older residents and those with disabilities; and increased pressure on public transport could strain cultural infrastructure such as bus stops and terminals near heritage sites. Possible solutions include expanding the public transport network (new routes, more bus stops, upgraded infrastructure), implementing traffic management measures (dedicated bus lanes, signal optimisation, congestion pricing), optimising resource allocation (public-private partnerships, technology for service optimisation), integrating cultural sensitivity into transport planning so that heritage sites remain well-connected, and engaging local communities and cultural institutions in transport development plans.

4.3.2 For Bangalore

Figure 12 shows the bubble plot of PTAL score and 2022 population estimate for Bangalore, and Figure 13 shows the fitted regression line.

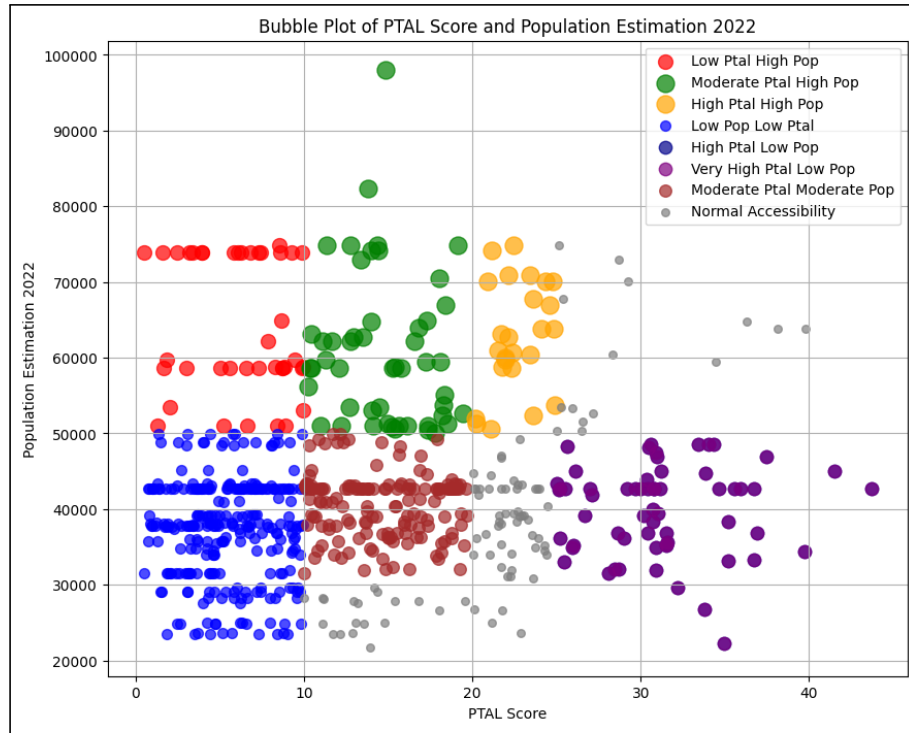


Figure 12: Bubble plot of PTAL score and 2022 population estimate, Bangalore.

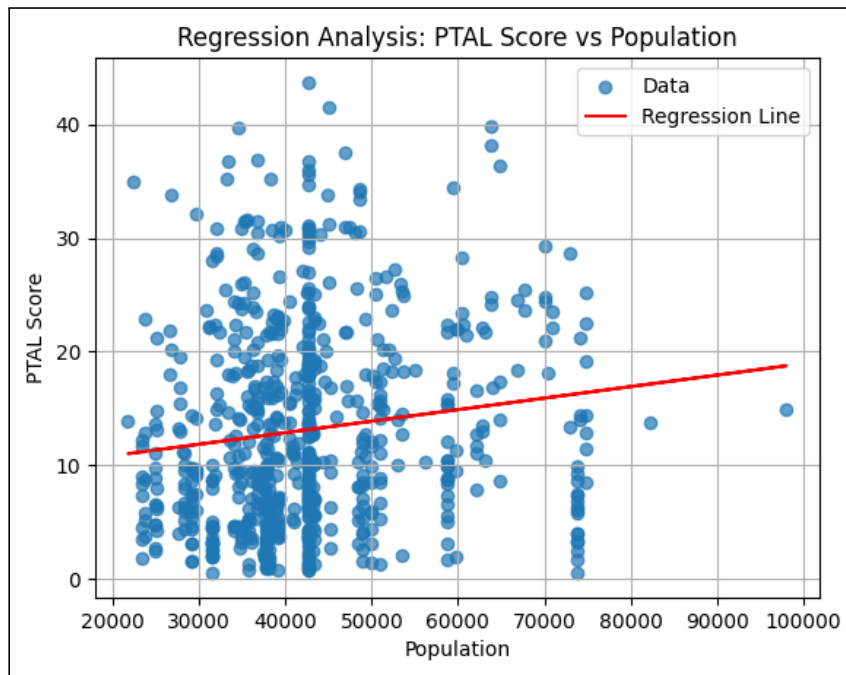


Figure 13: Regression analysis of PTAL score vs. population, Bangalore.

The regression analysis between population estimation and PTAL score in Bangalore revealed a statistically significant but weak *positive* relationship, suggesting that an increase in population tends to lead to a slight rise in PTAL score. This finding indicates that Bangalore's growing population could potentially benefit from improved public transport accessibility, attributable to infrastructure development (a corresponding expansion of public transport infrastructure, such

as new bus routes, metro lines, and bus rapid transit corridors, leading to a more extensive and accessible network), policy initiatives (subsidies for bus fares, dedicated bus lanes, and integrated ticketing systems encouraging more residents to switch to public transport), and technological advancements (real-time bus tracking apps, mobile ticketing, and intelligent traffic management systems improving travel planning and reducing waiting times).

Improved PTAL in Bangalore has several implications for the business environment: enhanced employee mobility through a well-connected public transport system reducing commute time; reduced traffic congestion easing the movement of goods and services; access to a wider talent pool as improved accessibility allows businesses to draw from a larger and more diverse workforce; and support for sustainable urban development by reducing reliance on private vehicles. Possible strategies to further enhance PTAL in Bangalore include continued infrastructure expansion (prioritising high-demand business hubs and under-connected residential areas), integration with other modes (autos, rickshaws, bicycles for first- and last-mile connectivity), demand-responsive transport in areas of low density or fluctuating demand, data-driven decision-making to identify low-accessibility areas, and public-private partnerships.

4.4 Comparison of Regression Results

Table 5 summarises the regression results for the two cities.

Table 5: Comparison of regression results: Varanasi vs. Bangalore.

Feature	Varanasi	Bangalore
R-squared	0.086	0.018
Population coefficient	-0.0002 (p = 0.031)	0.0001 (p = 0.000441)
Intercept	8.3325	8.7965
F-statistic and p-value	4.888 (p = 0.0315)	12.47 (p = 0.000441)

Key insights. The direction of the relationship between population and PTAL score differs between the two cities: Varanasi has a weak negative relationship, suggesting that increasing population tends to lead to a slight decrease in PTAL score, whereas Bangalore exhibits a weak positive relationship, indicating that population growth is associated with a slight increase in PTAL score. The strength of the relationship also differs: in Varanasi, the R-squared value of 0.086 suggests that only about 8.6% of the variability in PTAL scores can be explained by population, whereas in Bangalore the R-squared value of 0.018 is even lower, indicating population alone explains only about 1.8% of the variability. The significance of the population coefficient also differs: in Varanasi, the p-value of 0.031 suggests the negative relationship is statistically significant at the 5% level, while in Bangalore the p-value of 0.000441 indicates even more robust statistical significance for the positive relationship.

Possible explanations for the differences. Bangalore has invested more heavily in public transport infrastructure, such as metro lines, bus rapid transit corridors, and dedicated bus lanes, compared to Varanasi, likely contributing to the positive population-PTAL relationship.

observed there. Bangalore has also implemented various policies to promote public transport usage, including subsidies, integrated ticketing systems, and priority lanes, further improving accessibility as the population grows. Bangalore’s urban planning and transportation master plans have emphasised the development of a comprehensive and interconnected public transport network, while Varanasi’s urban planning may not have prioritised public transport to the same extent. Finally, Bangalore’s economic growth and vibrant business environment have driven investment in infrastructure and policy initiatives supporting accessibility.

5 Conclusion

The comparative analysis of PTAL scores in Bangalore and Varanasi highlights the importance of considering factors beyond population growth alone when assessing and improving public transport accessibility. Cities like Varanasi can learn from Bangalore’s experience by prioritising public transport infrastructure, implementing supportive policies, and adopting comprehensive urban planning strategies that promote a well-connected and accessible public transport system. By addressing their unique challenges and leveraging their respective strengths, both Varanasi and Bangalore can continue to enhance their PTAL scores, ensuring sustainable mobility, improved quality of life for residents, and a more attractive environment for businesses to thrive.

The grid-based PTAL mapping approach developed in this study, adapted from the London/TfL methodology to the data constraints and urban form of Indian cities, offers a quantitative, transferable measure of accessibility to public transport that can inform planning practice in several ways: formulating development/master plans with land use–transport integration, prioritising public transport and supporting investment in low-accessibility areas, formulating parking policies that discourage private vehicle use where PTAL is already high, and developing transit-oriented zoning regulations that concentrate density around high-PTAL nodes.

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